Time as a Dynamic Axis: An Operational Definition

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August 24, 2025

Time is not a backdrop; it is the counted advance of processes, and the universe sets the count.

Abstract

We define time as a dynamic axis: the process–distance advanced by an admissible clock, converted into a coordinate by the universal constant c. This operational definition, grounded in periodic physical transformations, recasts time not as a static backdrop but as an emergent coordinate from physical processes. We show that this definition leads naturally to the interpretation of time as a fourth dimension in relativistic spacetime.

1 Overview and Roadmap

Thesis at a glance. Time is a *dynamic axis*: the process–distance advanced by an admissible clock, converted into a coordinate by the universal constant c,

$$\Delta t^{(B)} = \frac{\Delta s_B}{c} = N_B\,\tau_B, \qquad \Delta s_B := N_B\,\lambda_B, \quad c = \frac{\lambda_B}{\tau_B}.$$

Logical steps.

1. **Operational stance.** Choose any *admissible* clock B: a system with stable, reproducible cycles (τ_B) and a calibrable scale λ_B satisfying $c = \lambda_B/\tau_B$. Define its *process–distance* $\Delta s_B := N_B\lambda_B$.

2. **Duration (mechanism–independent).** For any process window, assign the duration $\Delta t^{(B)} := N_B \tau_B = \Delta s_B / c$. This is what clocks measure—no background time assumed.

3. **Universality of c.** Mild symmetry (relativity principle, spatial homogeneity/isotropy, reciprocity) implies a single conversion constant c for all admissible clocks; hence the duration is independent of the clock mechanism up to a fixed calibration [4, 5, 6].

4. **Dynamic axis.** Because $\Delta t^{(B)}$ is defined from a clock's internal advance and is independent of spatial coordinates, it serves as an *independent* coordinate axis. Kinematic quantities are relational, e.g.

$$\mathbf{v}^{(B)} = \frac{d\mathbf{x}}{dt^{(B)}} = c\,\frac{d\mathbf{x}}{ds_B}.$$

5. **4D representation (in inertial regions).** With invariant two-way c, inertial charts are related by Lorentz transformations and admit a local 3+1 representation $(ct^{(B)}, \mathbf{x})$ with invariant

$$c^2\, d\tau^2 = c^2\,(dt^{(B)})^2 - d\mathbf{x}^{\,2},$$

   i.e., a Lorentzian metric with signature $(+,-,-,-)$ [7, 16].

6. **Scope and tests.** The construction matches standard kinematics (e.g., time dilation via light clocks) and metrology (optical clocks), extends locally to curved spacetime via $d\tau^2 = (1/c^2) g_{\mu\nu} dx^\mu dx^\nu$, and remains purely operational [15].

What we do *not* assume. We do not assume a pre-existing global time, discreteness of time, or any special properties of light beyond the operational role of the invariant c. For non-wave mechanisms, $\lambda_B := c/\nu_B$ is an *associated* calibration scale.

How to read this note. The next section formalizes admissible clocks and calibration; the following section defines duration and states the relational principle; then we introduce "time as a dynamic axis" and its 4D Lorentzian representation. Short optional modules (radar synchronization, light-clock dilation, invariant interval, clock equivalence) and a quantum Ramsey appendix can be read independently.

2 Admissible Clocks

Definition. A *clock* B is a system undergoing a reproducible periodic transformation $\mathcal{T}_{\text{clock}}$ with a well-defined period τ_B and an associated wavelength λ_B satisfying

$$c = \frac{\lambda_B}{\tau_B} \tag{1}$$

in the clock's rest frame; i.e., the standard wave relation $c = \lambda\,\nu$ with $\nu_B = 1/\tau_B$. *For mechanisms that are not intrinsically wave-based,* $\lambda_B := c/\nu_B$ is an *associated* scale introduced for calibration; it need not correspond to a radiated wavelength.

Admissibility requires:

(i) **Periodicity**: stable, repeatable cycles;

(ii) **Reproducibility under transport**: behavior remains consistent when transported between locations and inertial frames (as realized, e.g., by modern transportable and optical clocks [15]);

(iii) **Calibrability**: can be referenced to the universal relation (1).

Universality. We assume that all admissible clocks share the same conversion constant c (derivable from the relativity principle, spatial isotropy, and reciprocity in fuller treatments) [4, 5, 6].

Terminology. A *tick* denotes one completed cycle of an admissible clock. "Tick Theory" does *not* assume time is fundamentally discrete; it is an operational counting procedure fully compatible with continuous limits.

Gloss (process–distance). For later use, define the clock's *process–distance* over a window by $\Delta s_B := N_B\,\lambda_B$; the operational duration relative to B is $\Delta t^{(B)} = \Delta s_B/c = N_B\,\tau_B$.

3 Operational Definition of Duration

Primary definition. Let $N_B(\mathcal{T}_A)$ be the number of completed cycles of clock B during the process $\mathcal{T}_A$. The duration relative to B is

$$\Delta t_A^{(B)} := N_B(\mathcal{T}_A)\,\tau_B, \tag{2}$$

an explicitly operational stance ("time is what clocks measure") in the spirit of Einstein's 1905 construction of timekeeping and simultaneity [7].

Wave-calibrated representation (optional). When a wave-based calibration is chosen for B (with wavelength λ_B satisfying $c = \lambda_B/\tau_B$), one may equivalently write

$$\Delta t_A^{(B)} \;=\; \frac{N_B(\mathcal{T}_A)\,\lambda_B}{c} \;=\; \frac{\Delta s_B(\mathcal{T}_A)}{c}, \tag{3}$$

where $\Delta s_B := N_B\,\lambda_B$ is the clock's wave advance over the same window. This is just the standard wave relation $c = \lambda\nu$ with $\nu = 1/\tau_B$.[1]

Note (associated wavelength). For clocks that do not involve a propagating wave, $\lambda_B := c/\nu_B$ is a *defined associated scale* chosen so that $c = \lambda_B/\tau_B$; it need not represent a physical wavelength of the mechanism.

Principle (Relational time)**.** *Only relations between processes are measurable. Let B be any* admissible *clock that co-records a local process segment $\mathcal{T}$ along a worldline (i.e., is co-located on $\mathcal{T}$). Define*

$$T[\mathcal{T}] \;\equiv\; \Delta t^{(B)}(\mathcal{T}) \;:=\; N_B(\mathcal{T})\,\tau_B \;=\; \frac{\Delta s_B(\mathcal{T})}{c}, \qquad \Delta s_B := N_B\,\lambda_B.$$

Then T is additive under concatenation and, by the universality assumption on c (derivable from the relativity principle, spatial isotropy, and reciprocity [4, 5, 6]), mechanism-independent up to a fixed global scale (choice of units). In inertial regions of Minkowski spacetime—using the standard two-way invariance of c and the clock hypothesis—the functional $T[\mathcal{T}]$ coincides with proper time along that worldline [7, 16].

[1] For non-radiating clocks (mechanical, orbital, or internal atomic transitions), the operational definition (2) is sufficient and preferred. Equation (3) is a *representation* convenient when comparing to spatial measures; introducing an associated $\lambda_B = c/\nu_B$ is a metrological convention, not an additional physical claim. Modern optical clock practice exemplifies the required stability and calibrability [15].

Equation (3) restates this operationally: the duration of a process is the process–distance advanced by the clock's associated wave during that process, divided by the universal conversion constant c. The unit choice is fixed by selecting a particular standard clock (e.g., SI second).

Physical meaning. Here Δs_B is the cumulative internal advance of the clock's periodic process. For wave clocks, one tick advances the phase by 2π and the pattern by one wavelength, so over N_B ticks $\Delta s_B = N_B \lambda_B$. For non-wave (non-radiating) clocks we introduce an *associated* length per tick $\lambda_B := c\,\tau_B = c/\nu_B$ by metrological convention; this keeps $\Delta t^{(B)} = \Delta s_B/c = N_B \tau_B$ uniform across mechanisms. Note that Δs_B is neither a laboratory spatial displacement nor the Minkowski interval.

Example (Non-wave clock: pendulum). Consider a simple pendulum: a non-radiating, mechanical clock. In a uniform gravitational field and for small amplitudes, its period and frequency are

$$\tau_B = 2\pi\sqrt{\frac{L}{g}}, \qquad \nu_B = \frac{1}{\tau_B}.$$

Define its *associated wavelength* by the metrological convention

$$\lambda_B := \frac{c}{\nu_B} = c\,\tau_B,$$

which is a defined scale (not a physical wavelength of emitted radiation). For N_B complete swings, the operational duration is

$$\Delta t^{(B)} = N_B\,\tau_B \;=\; \frac{N_B\,\lambda_B}{c} \;=\; \frac{\Delta s_B}{c}, \qquad \Delta s_B := N_B\,\lambda_B.$$

Thus, the identity $\Delta t^{(B)} = \Delta s_B/c$ holds for a non-wave mechanism as well, illustrating *mechanism-independence. Admissibility caveat:* a simple pendulum is only *locally* admissible - τ_B depends on g (and weakly on amplitude beyond the small-angle regime), so reproducibility under transport fails when g changes; this is why atomic clocks are preferred as portable standards.

4 Time as a Dynamic Axis

Definition. Time is the *process–distance* advanced by any admissible reference system (clock), converted into a coordinate by the universal constant c [4, 5, 6]. For a clock B with period τ_B and (possibly associated) wavelength λ_B calibrated by $c = \lambda_B/\tau_B$, counting N_B ticks during a process gives

$$\Delta t^{(B)} \;=\; N_B\,\tau_B \;=\; \frac{N_B\,\lambda_B}{c} \;=\; \frac{\Delta s_B}{c}, \qquad \Delta s_B := N_B\,\lambda_B, \tag{4}$$

i.e., the wave-calibrated form of the operational relation $c = \lambda\nu$ with $\nu = 1/\tau_B$ [17]. Thus, the time coordinate is generated by the advance of physical processes, rather than assumed as a static backdrop—and functions as a *dynamic axis* independent of the spatial coordinates [7].

Consequence. With the relativity principle, spatial homogeneity/isotropy, reciprocity, and a universal c, $\Delta t^{(B)} = \Delta s_B/c$ serves as an independent coordinate, so events admit a local 3+1 chart $(ct^{(B)}, \mathbf{x})$ with Lorentzian structure [8, 16].

This dynamic axis is functionally distinct from spatial dimensions: it indexes the unfolding of causal processes and is represented by the time-like coordinate in a Lorentzian manifold.

No-Bridge Principle (Relational Reachability)

Principle (Principle (No-bridge, relational form))**.** *Motion is not the enumeration of intermediate points but a* correlation *between an evolving system A and an* admissible *clock B. If per tick of B the monitored quantity of A exhibits a strictly positive average advance, then a* finite *tick budget reaches the target. No continuum needs to be "crossed".*

Reachability lemma (operational). Let x be the coordinate of interest for system A. Suppose there exist $\delta > 0$ and an integer $m \geq 1$ such that for every consecutive block of m ticks of clock B, the net advance satisfies

$$\Delta x_{\text{block}} \;\geq\; \delta.$$

Then, starting from $x = 1$, one reaches $x = 2$ within at most

$$N \;\leq\; \left\lceil \tfrac{2-1}{\delta} \right\rceil m$$

ticks of B. In particular, the required number of ticks is finite.
Proof sketch. Concatenate blocks until the summed advance exceeds 1. Each block contributes at least δ, so at most $\lceil 1/\delta \rceil$ blocks are needed, i.e. at most $\lceil 1/\delta \rceil\, m$ ticks.

Corollary (dynamic-axis form). If along the evolution one has a bounded-away-from-zero relational drift

$$\frac{dx}{dt^{(B)}} \;\geq\; v_{\min} \;>\; 0 \quad \text{for a.e. ticks,}$$

then the dynamic-axis duration to cover Δx obeys the finite bound

$$\Delta t^{(B)} \;\leq\; \frac{\Delta x}{v_{\min}}.$$

Equivalently, in tick language: a positive average advance per tick implies a finite tick count to reach the target.

Concrete mappings (physics, not enumeration).

- **Spaceflight (Earth → Moon).** Let r be the Earth–craft range and B an onboard clock. If $\frac{dr}{dt^{(B)}} \geq v_{\min} > 0$ almost everywhere, the tick budget to traverse Δr is bounded by

$$\Delta t^{(B)} \;\leq\; \frac{\Delta r}{v_{\min}} \quad \Rightarrow \quad N_B \;\leq\; \frac{\Delta r}{v_{\min}\,\tau_B}.$$

- **Quantum evolution (two-level drive).** With Rabi frequency Ω, the phase advances as $\phi = \Omega\,\Delta t^{(B)}$. A π-pulse reaches the target in $\Delta t^{(B)} = \pi/\Omega$ (finite ticks $N_B = \Omega\,\Delta t^{(B)}/2\pi = \frac{1}{2}$).

- **Relativistic worldlines.** Between two events,

$$\Delta\tau = \frac{1}{c}\int \sqrt{c^2\,(dt^{(B)})^2 - d\mathbf{x}^{\,2}}\;.$$

  which is finite for physical paths (bounded speeds, finite segments). Distinct paths yield different finite $\Delta\tau$ without any "enumeration" of intermediate reals.

One-line takeaway. A strictly positive relational drift (per tick or per $\Delta t^{(B)}$) yields a *finite* tick budget to go from 1 to 2: motion is realized by *relations* and finite counts, not by visiting an infinity of intermediate points.

5 Time as the Fourth Coordinate Dimension

This framework shows that time behaves as a coordinate axis on equal mathematical footing with spatial dimensions. The process distance Δs_B is analogous to spatial displacements, and its conversion through c ensures compatibility with the relativistic spacetime structure [8, 16]. The Lorentzian metric's signature encodes the asymmetry between time and space, so this 'fourth dimension' is independent but not identical to the spatial axes [16].

> **Proposition.** *Time functions as the fourth coordinate dimension in a relativistic manifold, generated operationally by periodic processes and scaled by the invariant conversion constant c [8].*

Unlike space, however, time emerges from the evolution of processes and may retain a directional or causal asymmetry depending on the context (e.g., thermodynamics). Nonetheless, geometrically and operationally, it fulfills the role of a fourth dimension in Minkowski-type spacetimes [8].

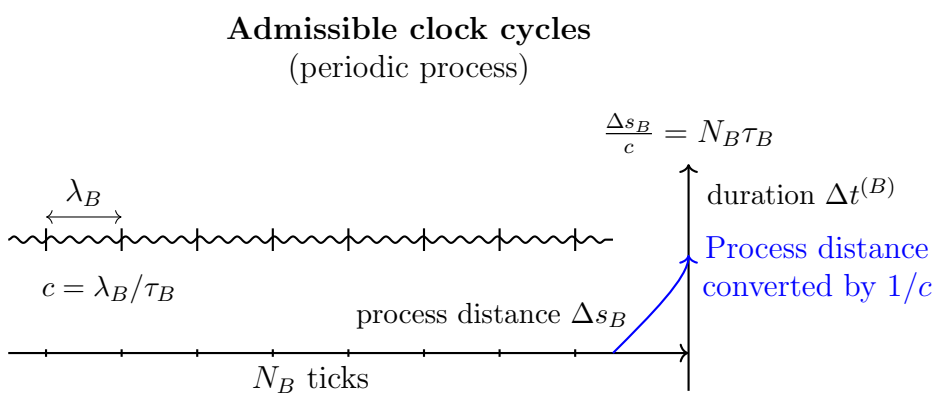

Figure 1: An admissible clock defines time operationally. Counting N_B cycles gives a process–distance $\Delta s_B = N_B \lambda_B$; dividing by the universal conversion constant $c = \lambda_B/\tau_B$ yields the duration $\Delta t^{(B)} = \Delta s_B/c = N_B \tau_B$.

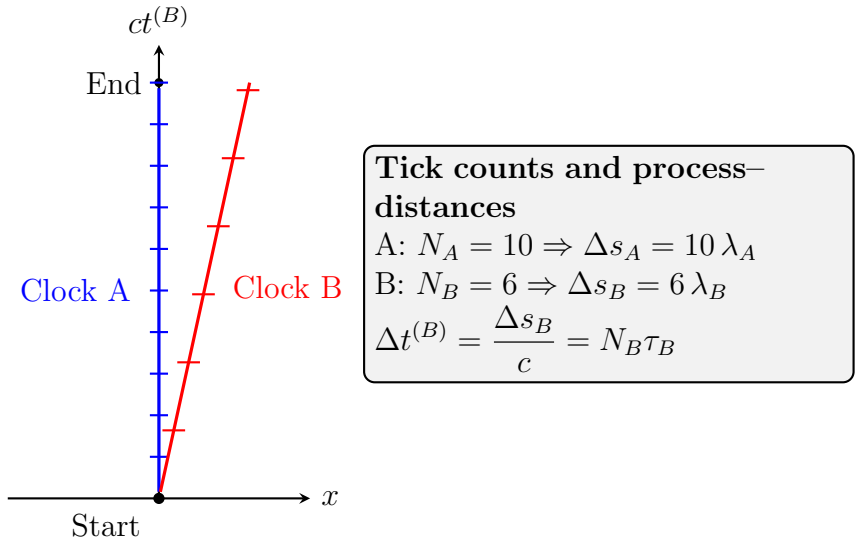

Figure 2: Two admissible clocks connecting the same start/end events. Clock A at rest (blue) accumulates $N_A = 10$ ticks; the moving clock B (red) accumulates $N_B = 6$. Since $\Delta t^{(B)} = \Delta s_B/c = N_B \tau_B$, fewer ticks mean a smaller elapsed duration for B - an operational depiction of time dilation.

6 Future Additions

The present note is intentionally minimal: it defines duration operationally and frames time as a dynamic axis generated by admissible clocks. To keep the scope lean while signposting next steps, we list compact add-ons that can be included as brief modules or appendices without altering the core construction. In all items below, time is treated as an independent coordinate $t^{(B)}$ derived from process–distance ($dt^{(B)} = ds_B/c$; over a window $\Delta t^{(B)} = \Delta s_B/c$) and paired with spatial coordinates $\mathbf{x}$. Here, ds_B denotes the differential process–distance of the chosen admissible clock B.

(A) Frame-level time via radar (Einstein synchronization). Using invariant two-way speed c, define event coordinates on an inertial frame whose *spatial origin is co-located with the admissible clock B*. If a light signal is sent at the frame's clock reading $t_{\mathrm{s}}^{(B)}$ and received back at $t_{\mathrm{r}}^{(B)}$, assign to the

reflection event P:

$$t_E^{(B)}(P) = \tfrac{1}{2}\big(t_{\mathrm{s}}^{(B)} + t_{\mathrm{r}}^{(B)}\big), \qquad x_E^{\parallel}(P) = \tfrac{c}{2}\big(t_{\mathrm{r}}^{(B)} - t_{\mathrm{s}}^{(B)}\big),$$

with $x_E^{\parallel}$ along the line of sight and transverse coordinates set by a rigid orthogonal lattice (or two-baseline radar). This promotes local durations to a frame time field $t^{(B)}(\mathbf{x})$ without altering the dynamic-axis meaning of time [7].

(B) Nontrivial kinematics from a light clock (time dilation). For a transverse light clock of arm length L moving at speed v relative to an inertial frame with time $t^{(B)}$, the path relation over one tick gives

$$\left(\tfrac{c\,\Delta t^{(B)}}{2}\right)^2 = \left(\tfrac{v\,\Delta t^{(B)}}{2}\right)^2 + L^2 \;\Rightarrow\; \Delta t^{(B)} = \gamma\,\Delta\tau_0, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

$$\Delta\tau_0 = \frac{2L}{c}.$$

Here $\Delta\tau_0$ is the proper tick of the co-moving light clock, while $\Delta t^{(B)}$ is the coordinate duration recorded by the inertial frame's time $t^{(B)}$ [16].

(C) Invariant interval and proper time (geometric tie-in). In inertial charts with time coordinate $x^0 = ct^{(B)}$, one has the quadratic invariant

$$c^2(\Delta\tau)^2 \;=\; c^2\big(\Delta t^{(B)}\big)^2 - \|\Delta\mathbf{x}\|^2,$$

consistent with metric signature $(+,-,-,-)$. Ideal admissible clocks measure $\Delta\tau$ along worldlines, specifying precisely how time is an independent axis with distinct signature [16].

(D) Clock equivalence on a worldline (formal lemma). For admissible clocks B, B' co-located on the same process segment $\mathcal{T}$,

$$T_{B'}(\mathcal{T}) = K_{BB'}\,T_B(\mathcal{T}), \qquad K_{BB'} > 0,$$

hence equality after a single calibration ($K_{BB'} = 1$). This follows from additivity and stability (optionally, an Archimedean/continuity property), with no additional physics assumptions [2].

(E) Empirical anchors (metrology). Standard tests align with the construction: muon lifetime dilation, optical-clock redshift in weak fields, and transport/synchronization comparisons [13, 15, 11, 10]. Clarify admissibility limits (clock hypothesis; environmental sensitivities of non-ideal mechanisms)

(F) Beyond inertial regions (brief note). Using coordinates $x^0 = ct^{(B)}$, proper time in curved spacetime generalizes via

$$d\tau^2 = \frac{1}{c^2}\, g_{\mu\nu}\, dx^\mu dx^\nu$$

for signature $(+, -, -, -)$. Local inertial frames recover the inertial formulas with $t^{(B)}$ as the dynamic axis.

(G) Quantum interface (deferred). Phase accumulation and interferometric clocks couple directly to the dynamic axis, e.g. $\phi = \omega\, \Delta t^{(B)}$ with $\omega = 2\pi\nu_B$, while keeping the present note purely operational and relativistic [18].

7 Conclusion

We have presented a minimal, operational construction of time: the duration of a process is defined by counting ticks of an admissible clock, $\Delta t^{(B)} = N_B \tau_B$, with an equivalent wave-calibrated form $\Delta t^{(B)} = \Delta s_B / c$ where $\Delta s_B = N_B \lambda_B$ and $c = \lambda_B / \tau_B$. This identifies time a *dynamic axis* increase generated by the advance of physical processes rather than as a static backdrop.

Within inertial regions, and adopting the standard two-way invariance of c [7], this local construction supports the usual relativistic interpretation: time functions as an independent coordinate on equal mathematical footing with space (with a distinct metric signature), enabling a local 3+1 chart $(ct^{(B)}, \mathbf{x})$. The "future additions" outline short, optional modules, radar synchronization, light-clock dilation, the invariant interval, and a clock-equivalence statement of a lemma, which elevate the presentation from operational foundations to the familiar Lorentz–Minkowski structure without altering the core thesis [16, 8].

The guiding perspective remains relational: only comparisons between processes are measurable, and the universe provides the references. In this view, time is precisely the counted advance of admissible processes, scaled by c [2].

Appendix: A Minimal Quantum Clock (Ramsey) and the Relation $\phi = \omega_B \tau$

A. Model of an admissible quantum clock

Consider a two–level system B with ground $|g\rangle$, excited $|e\rangle$, and rest–frame Hamiltonian $H_B = \frac{\hbar\omega_B}{2}\sigma_z$ (transition angular frequency $\omega_B > 0$). Let the clock co-move on a worldline γ with proper-time parameter τ. The free evolution is

$$U_0(\tau) = e^{-\frac{i}{\hbar}H_B\,\tau} = \exp\Big(-\,\tfrac{i\omega_B\tau}{2}\sigma_z\Big),$$

so any superposition accumulates a relative phase at rate ω_B *per unit proper time* (τ). [18]

B. Ramsey sequence (operational readout of τ)

Prepare $|g\rangle$. Apply a resonant $\pi/2$ pulse to create $|\psi(0)\rangle = \frac{1}{\sqrt{2}}(|g\rangle + |e\rangle)$. Allow free evolution for a co-moving proper duration τ, then apply a second $\pi/2$ pulse with analysis phase θ. The excited-state probability is

$$P_e(\tau;\theta) = \tfrac{1}{2}\Big[1 + C(\tau)\cos\big(\omega_B\tau + \phi_0 - \theta\big)\Big],$$

where ϕ_0 is a preparation phase and $C(\tau) \in (0, 1]$ the Ramsey contrast (ideal clock: $C = 1$). Thus the *measured* phase advance is

$$\phi_B(\tau) = \omega_B\tau + \phi_0 \quad \Rightarrow \quad \boxed{\phi_B(\tau) - \phi_0 = \omega_B\,\tau},$$

with maximal slope at $\theta = \phi_0 + \frac{\pi}{2}$. [18]

C. Identification with the dynamic axis

For a co-moving admissible clock, the operational time in the main text satisfies

$$\Delta t^{(B)} = \frac{\Delta s_B}{c} = N_B\,\tau_B = \frac{\phi_B}{\omega_B}, \qquad \tau_B = \frac{2\pi}{\omega_B}, \quad N_B = \frac{\phi_B}{2\pi}.$$

Hence, along the clock's worldline γ,

$$\boxed{\Delta t^{(B)} = \tau} \qquad \text{and} \qquad \boxed{\phi_B = \omega_B\,\Delta t^{(B)} = \omega_B\,\tau}.$$

This realizes the "dynamic axis" in quantum terms: the process–distance is the accumulated phase divided by ω_B, consistent with $\Delta t^{(B)} = \Delta s_B/c$ using $\lambda_B = c/\nu_B$ and $\omega_B = 2\pi\nu_B$.

D. Relativistic tests (two clocks or two paths)

Let two identical quantum clocks traverse worldlines γ_1, γ_2 with proper times τ_1, τ_2. Their differential readout is

$$\Delta\phi = \phi_1 - \phi_2 = \omega_B(\tau_1 - \tau_2).$$

In flat spacetime at constant speeds v_1, v_2 relative to an inertial frame, $\Delta\tau = \Delta t^{(B)}\sqrt{1 - \frac{v^2}{c^2}} = \gamma^{-1}\,\Delta t^{(B)}$, so

$$\Delta\phi = \omega_B\, t^{(B)}\big(\gamma_1^{-1} - \gamma_2^{-1}\big), \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

In a weak gravitational field with potential Φ, the fractional redshift $\Delta f/f = \Delta\Phi/c^2$ gives

$$\Delta\phi = \omega_B \int \frac{\Delta\Phi}{c^2}\, dt^{(B)},$$

recovering gravitational time-dilation in the Ramsey phase. [22]

E. Measurement bounds (what limits $\Delta t^{(B)}$ estimation?)

Projection/shot noise (standard quantum limit). For M uncorrelated repetitions (or M atoms), the Fisher information for $P_e = \frac{1}{2}[1 + C\cos\phi]$ at optimal bias ($\phi \approx \pi/2$) is $F_1 = C^2$, so the Cramér–Rao bound is

$$\Delta\phi \;\geq\; \frac{1}{C\sqrt{M}} \quad \Rightarrow \quad \Delta t^{(B)} \;\geq\; \frac{1}{\omega_B\, C\sqrt{M}}.$$

Dephasing $C(\tau) = e^{-\chi(\tau)}$ degrades precision accordingly. [19]

Quantum speed limits (MT/ML). Writing $\theta = \arccos|\,\langle\psi(0)|\psi(\tau)\rangle\,|$, the evolution time obeys

$$\tau \;\geq\; \max\left\{\frac{\hbar\,\theta}{2\,\Delta E},\; \frac{\pi\hbar}{2\,(\langle H\rangle - E_0)}\right\},$$

(Mandelstam–Tamm / Margolus–Levitin), which for free Ramsey precession is of order $1/\omega_B$. [20, 21]

Systematic limits. Finite interrogation time, dead time, reference laser noise, and environment sensitivity set practical bounds; these affect $C(\tau)$ and the total time uncertainty but do not alter $\phi = \omega_B \tau$.

F. Admissibility and mechanism–independence

Any two admissible quantum clocks B, B' co-moving on the same worldline yield estimates related by a fixed calibration factor:

$$\hat{t}^{(B')} = K_{BB'}\, \hat{t}^{(B)}, \qquad K_{BB'} = \frac{\omega_B}{\omega_{B'}}.$$

After one calibration ($K_{BB'} = 1$), both realize the same τ within quantum uncertainties. Thus the operational time of the main text coincides with proper time even for intrinsically quantum clocks.

G. Summary

A minimal Ramsey clock realizes the dynamic axis as follows: (i) free evolution imprints $\phi = \omega_B \tau$; (ii) the estimator $\hat{t}^{(B)} = \hat{\phi}/\omega_B$ returns proper time along the clock's worldline; (iii) precision is bounded by projection noise and decoherence, not by any mismatch between SR and QM. Differential phases between distinct worldlines measure $\Delta\tau$, directly connecting the operational definition of time to relativistic kinematics.

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